

PAPER • OPEN ACCESS

Software system for automatic reading, storing, and evaluating scanned paper Evaluation Sheets for questions with the choice of one correct answer from several offered

To cite this article: Đulaga Hadžić *et al* 2023 *IOP Conf. Ser.: Mater. Sci. Eng.* **1298** 012020

View the [article online](#) for updates and enhancements.

You may also like

- [The fractional symmetric rigid rotor](#)
Richard Herrmann
- [New parameters of the deformed Woods-Saxon potential for \$A=110-210\$ nuclei](#)
J Dudek and T Werner
- [Physical processes in the solar atmosphere associated with flares](#)
B V Somov and S I Syrovatski

Software system for automatic reading, storing, and evaluating scanned paper Evaluation Sheets for questions with the choice of one correct answer from several offered

Dulaga Hadžić¹, Enes Saletović¹ and Zinaid Kapić²

¹Polytechnic Faculty, University of Zenica, Zenica, Bosnia and Herzegovina

²Faculty of Technical Engineering, University of Bihać, Bihać, Bosnia and Herzegovina

E-mail: djulaga.hadzic@unze.ba

Abstract. Although the technology to automatically score multiple-choice tests has been around for several decades, it is still not as widely available or affordable, especially for paper-based test processing. The main reasons that hinder these processes are the lack of software systems capable of working with contents that are not filled optimally and do not require expensive scanners or other costly equipment. In this study, we present a software system for the automatic reading, storing, and evaluation of scanned assessment sheets. This software system offers a solution that only requires the usual scanned Evaluation Sheets in the form of a not high-resolution image and with the help of specific markers, the system performs reading, storage, and scoring. The user interface is designed to read and display the data from the Evaluation Sheets in detail so that they are very understandable to the user and allow him to quickly spot any errors. The tool has been validated over six years of use and has been continuously improved throughout that period. Thus, this software system achieves a high level of reliability and sensitivity to different levels of the quality of filling out Evaluation Sheets by students.

1. Introduction

A traditional multiple-choice question (MCQ) is a type of question in which a student selects one answer from several choices provided [1]. However, implementing online computer-assisted assessment (CAA) can be challenging due to the high cost of setup and maintenance. Multiple-choice tests are used in important examinations to determine students' academic excellence [2]. Multiple-choice questions (MCQ) tests are considered a direct form of written assessment [3]. Despite the adaptability of other alternatives to assessment techniques, such as modified essay questions, the MCQ is still considered the most reliable technique to avoid any misinterpretation of answers [4, 5].

In Bosnia and Herzegovina, specifically in Tuzla Canton, at the end of the school year, the graduation exam is held for all students of both primary and secondary schools. As part of the matriculation examination, the Integral test is taken, which contains a certain number of questions with four choices for each question, from which they can select only one answer. The Integral test contains questions over several areas (subjects). Grading or evaluation of completed tests is done percentage-ally (from 0 to 100%) for the entire test as well as for individual areas (subjects). In further work, as a case study, we will analyze the Integral test for elementary school students, which is conducted in paper form. Considering the large number of students taking the Integral test at the same time (approximately 4000 students), the need to automate the review and evaluation of completed tests arose. The integral test for elementary school students contains 60 questions, which means that it would be necessary in a very short time to manually review over 24,000 (60 questions x 4,000 students) questions and determine whether the students answered each of them correctly or incorrectly. Of course, this is only one part of



reviewing the tests. Additionally, it is necessary to calculate the overall test result for each student as well as the test results for individual areas (subjects) within that test.

At the beginning of the matriculation exam, the review of tests was conducted manually, which posed significant challenges due to the large number of participating students [6]. Thus, the need to implement a specific system that could automate the process of reviewing and evaluating tests without introducing any expensive additional IT equipment quickly emerged. Therefore, an Evaluation Sheet was added to the Integral Test, enabling students to transfer their answers from the test onto this sheet. In the following sections of this article, we will introduce an in-house software system specifically designed to process scanned evaluation sheets. This system efficiently analyzes the sheets and creates comprehensive results for students' tests from the Integral test. These results include both the overall test score and the individual area (subject) scores, presented in a percentage format.

2. Integral Test and Evaluation Sheet

As already mentioned, the Integral Test for elementary school students contains 60 questions from 14 different areas (subjects). The number of questions by specific areas (subjects) is shown in Table 1.

Table 1. Structure of the Integral Test

	Area (Subject)	Number of questions
1	Bosanski/Hrvatski/Srpski jezik i književnost (eng. Bosnian/Croatian/Serbian language and literatures)	5 (1-5)
2	Prvi strani jezik (eng. First foreign language)	5 (6-10)
3	Matematika (eng. Mathematics)	5 (11-15)
4	Fizika (eng. Physics)	5 (16-20)
5	Informatika (eng. Informatics)	5 (21-25)
6	Biologija (eng. Biology)	5 (26-30)
7	Hemija (eng. Chemistry)	5 (31-35)
8	Historija (eng. History)	5 (36-40)
9	Geografija (eng. Geography)	5 (41-45)
10	Tehnička kultura (eng. Technical culture)	3 (46-48)
11	Likovna kultura (eng. Art culture)	3 (49-51)
12	Muzička kultura (eng. Music culture)	3 (52-54)
13	Tjelesni i zdravstveni odgoj (eng. Physical and health education)	3 (55-57)
14	Građansko obrazovanje (eng. Citizenship education)	3 (58-60)

In the Integral test, each question is accompanied by four possible answers marked with A, B, C and D. Figure 1 shows the structure of the first question from the Integral Test held at the External Matura for elementary schools at the end of the 2021/2022 academic year.

• BOSANSKI, HRVATSKI, SRPSKI JEZIK I KNJIŽEVNOST		
RB	TEKST PITANJA:	
1.	Pažljivo pročitaj stihove narodne balade Smrt Omera i Merime: <i>Nije blago ni srebro ni zlato, Već je blago što je srcu drago. Vrsta stih a u ovoj pjesmi je:</i>	
	PONUĐENI ODGOVORI:	
	A	jedanaesterac
	B	deseterac
	C	dvanaesterac
	D	deveterac
	TAČAN ODGOVOR:	

Figure 1. Structure of questions in the Integral Test.

In Figure 1, the TAČAN ODGOVOR (CORRECT ANSWER) field is where the student marks the corresponding letter (A, B, C, or D) for the answer they believe is correct for each question. There are

60 questions with this structure in the Integral Test. An example of an Integral Test can be viewed at the following link.¹

Once the student has answered all the questions or the ones they believe they can answer correctly, they proceed to transfer their marked answers to the Evaluation Sheet. Figure 2 provides an example of the Evaluation Sheet.

The Evaluation Sheet is divided into four parts:

- First Part:** Contains a QR code and the student ID 2037900013630.
- Second Part:** Contains the language selection 'Engleski jezik'.
- Third Part:** Contains the title 'OCJENJIVAČKI LIST' and instructions for marking answers.
- Fourth Part:** Contains a grid of 60 questions, each with four options (A, B, C, D) and a bubble for marking the answer.

Figure 2. An example of a filled-in Evaluation Sheet.

The Evaluation Sheet consists of the following parts (Figure 2):

- The First Part contains the student's ID
- The Second Part contains the ID for the first foreign language
- The Third Part contains the ID for the test combination
- The Fourth Part is intended for the transfer of marked answers for questions (from 1 to 60) from the Integral Test to the Evaluation Sheet

The student's ID is 13 digits, the first 9 of which indicate the elementary school the student attends, and the last 4 digits represent the student's unique identifier in the entire system (it can appear only once and belong to one student). The student's ID is printed on the Evaluation Sheet in numerical form and in the form of a 2D barcode. An example of the student's ID written in the Evaluation Sheet in the previously mentioned two ways is shown in the First Part of Figure 3.

The student ID is printed in numerical form for the reason that the numerical form is comprehensible to humans and the student ID in the form of a 2D barcode is comprehensible to the software system that will process (evaluate) the Evaluation Sheets.

¹ http://cms1.pztz.ba/userfiles/malamatura/files/Testovi/Matura2022/MMatura_test_april_2022_en.pdf

In the second part of the Evaluation Sheet, there is an identification field for the student's first foreign language, which corresponds to the language they study in the final grades of elementary school. The software system can recognize two options for the first foreign language: English (Engleski jezik) and German (Njemački jezik). Figure 3 illustrates the available options that the software system can identify for this field.

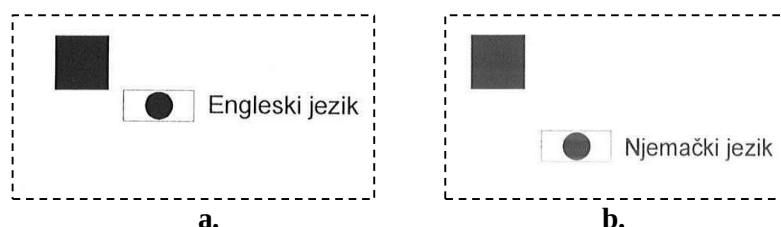


Figure 3. Examples for First Foreign Language ID.

In Figure 3, two Evaluation Sheets are shown. Under **a.** only English is shown as the first foreign language and on the second Evaluation Sheet under **b.** only German language is displayed as the first foreign language, but in different positions (coordinates). Based on these positions (coordinates), the software system determines a student's first foreign language and records it. On the basis of the first foreign language, at the end of the Integral Test, the corresponding Evaluation Sheet with the appropriate ID for the foreign language is printed.

The ID of the test combination is recorded in the Third Part of the Evaluation Sheet. Figure 2 shows an example of a combination label for one test. The number of possible combinations of the Integral Test is four. Based on the example shown in Figure 2, the ID of the combination of the Integral test is 1. This combination of the Integral Test is important from the aspect of the combination of correct answers for the questions that will be used when evaluating (reviewing) the completed test, as well as the Evaluation Sheet can be linked with the appropriate combination of the Integral Test.

In the Fourth Part of the Evaluation Sheet, there are the transferred selected answers from the Integral Test. Figure 2 shows the Fourth Part of the Evaluation Sheet, which consists of 3 columns with 20 IDs, one for each question. Each question ID is preceded by its ordinal number (RB). Also, the ID for one question contains four offered selection options that represent the transmitted answer from the Integral test for that question.

In Figure 4, below the title of the Evaluation Sheet (OCJENJIVAČKI LIST), there is a short instruction, i.e. examples of how to fill in (transmit) an answer to a specific question in the Evaluation Sheet and an example of how to cancel that answer and fill in (transmit) another answer for the same question.



Figure 4. An example of the correct selection of an answer, with an example of how that answer is canceled and another is selected.

For example, in Figure 4, in the first example, the answer to a specific question is marked under B, and in the second example, answer B is canceled and a new answer is marked (transferred) under C for the same question.

3. Software system with main functionalities

As part of this project, a software system for the automatic reading, storing, and evaluation of scanned Evaluation Sheets was developed. The software system was developed using .NET technology in the C# programming language as a desktop application. This system uses already externally scanned Evaluation Sheets as input parameters, and their reading and recording in the database can be done in two ways:

- Semi-automatic - reading of Evaluation Sheets is done one by one, where after one Evaluation Sheet has been read, it is stored in the database by clicking on the **Postavi** (Set) button.
- Automatic - reads and writes to the database successively a large number of Evaluation Sheets placed in one folder, which usually represents all Evaluation Sheets for students of one elementary school.

3.1. Semi-automatic reading

Below is a detailed description of the semi-automatic reading and recording of data from the Evaluation Sheets through the mentioned software system. Figure 5 shows the interface of the part of the software system through which the Semi-automatic reading of Evaluation Sheets is performed.

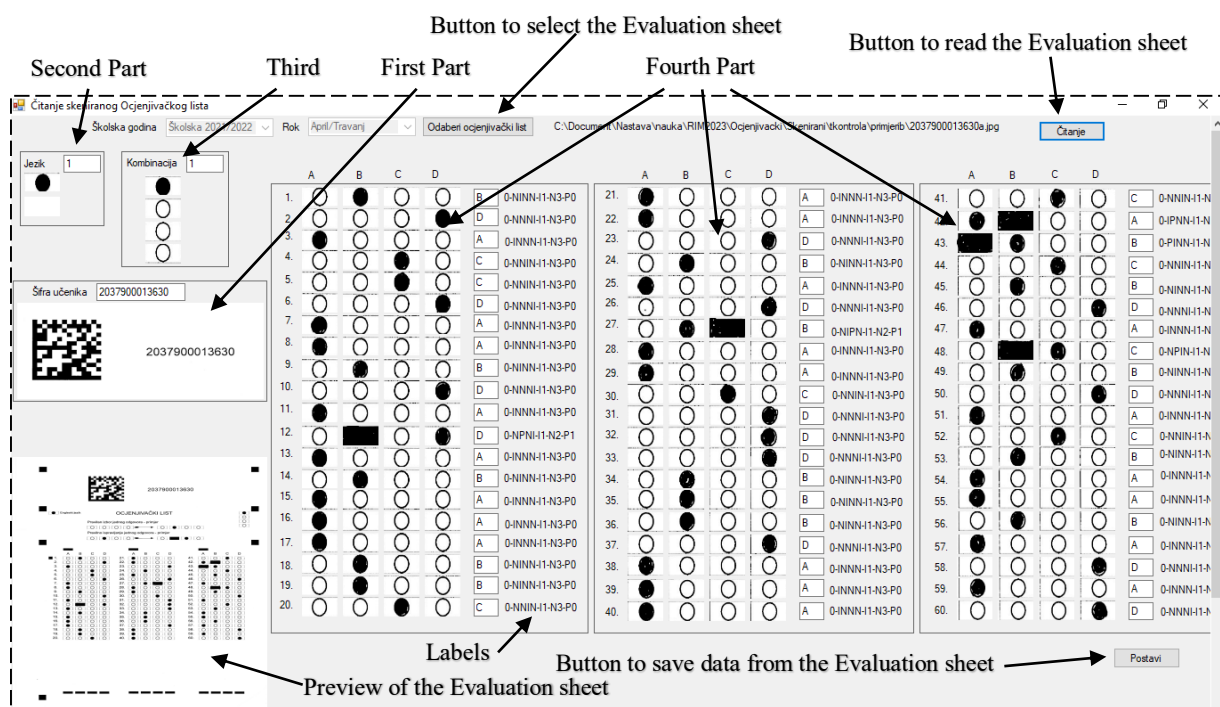


Figure 5. Interface for semi-automatic selection, reading, and storage of one scanned Evaluation Sheet

The user process a specific scanned Evaluation Sheet by clicking the "Odaberi ocjenjivački list" (Select evaluation sheet) button. Once selected, the Evaluation Sheet is read by clicking the "Čitaj" (Read) button and its content is populated in the corresponding fields of the interface shown in Figure 5. The interface design in Figure 5 aligns with the different parts of the Evaluation Sheet as shown in Figure 2. Figure 6 shows a more detailed perspective of the first, second, and third components of the Evaluation Sheet as seen in Figure 5.

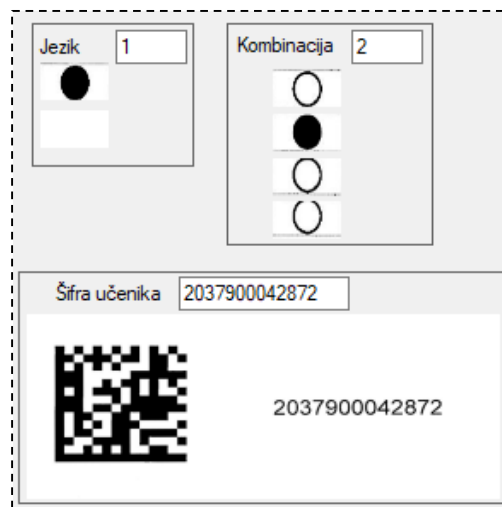


Figure 6. Read the first, second, and third parts of the Evaluation Sheet from Figure 2

Next, we will provide an overview of how the software system reads the Evaluation Sheet. Different ways of processing this type of question and answer are presented in the literature. One approach involves creating a template or mask that matches the layout of the Evaluation Sheet. This mask is then compared against the scanned student grade sheet, allowing the system to extract the answers (true or false) to the questions based on the detected markings.

Optical Mark Reading (OMR) is a technique used to automatically read user input in surveys, questionnaires, interviews, and testing systems [7]. OMR uses sensors to evaluate student responses by determining whether a predefined position is empty or already marked [8].

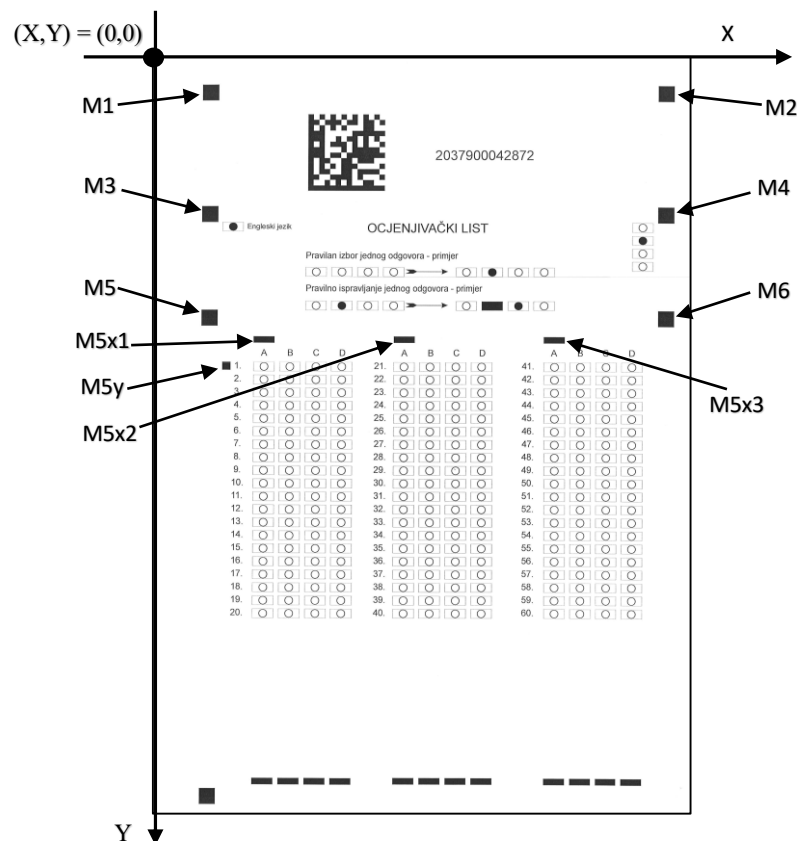


Figure 7. Markers and coordinate system for "movement" according to the Evaluation Sheet

The software system presented in this paper uses an optical mark reading approach. It "moves" around the Evaluation Sheet and reads individual contents. To move around the Evaluation Sheet, the software system uses the markers that are marked as $M1$ to $M7$ in Figure 7. Movement along the Evaluation Sheet starts from the point with coordinates $X = 0$ and $Y = 0$, located in the upper left corner of the Evaluation Sheet (Figure 7). The search is based on a function that works on the principle of comparing the colors of individual pixels. Hence, the software system will search all the pixels from point (0,0) to the first pixel that has a different color from the previous ones. In this case it will have a black color based on the previous pixels that are white. That pixel represents the top left corner of the $M1$ marker. The X and Y coordinates of the previously identified pixel are recorded for further calculations. The same methodology is used for all other markers in the Evaluation Sheet shown in Figure 7.

In the first step, after selecting the file with the Evaluation Sheet, the (X, Y) coordinates of the markers $M1$ and $M2$ are determined, and based on their positions, it is determined whether the Evaluation Sheet is "slanted", i.e. whether $Y_{M1} - Y_{M2} < > 0$, and if so, then the Evaluation Sheet is corrected (it is rotated by a certain angle to the left or right depending on whether the difference $Y_{M1} - Y_{M2}$ is less or greater than zero). Also, based on the distance between the markers $M1$ and $M2$, zooming (increased or reduced) is carried out according to the already defined standard dimensions of the Evaluation Sheet. After the previous corrections have been made, the Evaluation Sheet is saved in a separate file with a .jpeg extension for further processing.

In the second step, based on certain positions of the markers $M1$ (X_{M1}, Y_{M1}) and $M2$ (X_{M2}, Y_{M2}), the searching of the **First part** in the Evaluation Sheet begins (Figure 2). After the 2D barcode is found, it is read using the defined function, and the read value is placed in the corresponding field in the interface shown in Figure 5 (First part).

In the third step, the "movement" along the Evaluation Sheet continues and the $M3$ marker with coordinates (X_{M3}, Y_{M3}) is found, and based on it, the **Second Part** in the Evaluation Sheet from Figure 2 is searched and the ID of the first foreign language is determined through the corresponding function for reading this type of mark. and records in the appropriate field in the interface shown in Figure 5 (Second Part).

In the fourth step, the $M4$ marker with coordinates (X_{M4}, Y_{M4}) is found, and based on it, the **Third Part** is searched in the Evaluation Sheet from Figure 2 and through the appropriate function for reading this type of marker, the ID of the test combination is determined and recorded in the corresponding field in the interface shown on Figure 5 (Third Part).

In the fifth step, marker $M5$ with coordinates (X_{M5}, Y_{M5}), is found, and then from that point marker $M5y$ with coordinates (X_{M5y}, Y_{M5y}) is found. The point with coordinates (X_{M5y}, Y_{M5y}) contains the Y coordinate of the first mark in the first column of the Fourth Part of the Evaluation Sheet. By searching from marker $M5$ with coordinates (X_{M5}, Y_{M5}), marker $M5x1$ with coordinates (X_{5x1}, Y_{5y1}) is found, then marker $M5x2$ with coordinates (X_{5x2}, Y_{5y2}) and finally marker $M5x3$ with coordinates (X_{5x3}, Y_{5y3}). Recorded coordinates X_{5x1} , X_{5x2} and X_{5x3} from markers $M5x1$, $M5x2$ and $M5x3$ represent the value of X coordinates of the first mark in the first column, the first mark in the second column and the first mark in the third column, respectively. This means that the coordinates of the upper left corner of the listed first markers by column were found and recorded, namely (X_{M5x1}, Y_{M5y}) for the first, (X_{M5x2}, Y_{M5y}) for the second and (X_{M5x3}, Y_{M5y}) for the third column.

After the previous three points have been identified and recorded, and given that the width and height of one mark, the distance between the marks and the distance between the lines (questions) of the marks are already known, it is possible to separate each of the marks through certain loops, i.e. 240 of them (60 questions x 4 marks). For each question we have 4 marks representing 4 possible answers, where the first mark represents the answer under A, the second mark represents the answer under B, the third mark represents the answer under C and the fourth mark represents the answer under D.

The following describes how the software system reads the transmitted answers from the Evaluation Sheet. The software system defined three possible label states within a question, which are shown in Figure 8.

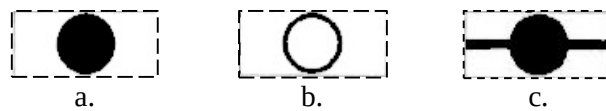


Figure 8. Three possible states of marks within one question: a. Izabran (Selected), b. Nije izabran (Not selected), and c. Poništen (Cancelled)

When analyzing the states of marks within a question, each of the four marks is compared with marks with three predefined states (Figure 8), and their similarities are calculated via the appropriate function.

The analyzed mark will represent the state with which it has the greatest similarity. The stated states under a., b., and c., in Figure 8 are recorded as I, N or P, respectively, in the Label located next to a specific question in the interface shown in Figure 5. In this way, in addition to recording the selected answer, it also records the answers which were selected and canceled as well as whether the student did not try to answer the given question at all or if he chose more than one offered answer. Based on the records stored on the label next to the marks for one question, the software system records in the corresponding field the answer chosen by the student (A, B, C, or D) or records the value 0 if the student did not offer an answer at all, or offered more than one answer.

Here it is necessary to further clarify the appearance of the mark for canceling the selected answer in Figure 8 since this mark does not have the same appearance as the mark for canceling the transferred answer shown in Figure 4. This was done for practical reasons, because the student will not always accurately cancel the marked answer, i.e. completely black out the answer mark, but it can be only partially. With that in mind, the response cancellation tag has the layout shown in Figure 8. Tag state under c. will always have the maximum similarity with the mark for canceling the answer, even in the case when it will not be completely filled in black.



Figure 9. Examples of reading and recording the selected answer for the questions

Figure 9 shows examples of reading labels for individual questions and how the answer was recorded in the interface shown in Figure 5.

Let's analyze question ordinal number 42 from Figure 9. The Label for this question is filled with the following content 0-IPNN-I1-N2-P1 which means the following:

IPNN: I – Selected, P – Cancelled, N – Not selected, and N – Not selected.

I1 – 1 mark is selected

N2 – 2 marks not selected

P1 – 1 mark canceled

In the field for recording the selected answer for question 42, the mark A is written, which represents the correct reading of the answer transferred to the Evaluation Sheet.

After all the data from the Evaluation Sheet have been read and recorded in the interface shown in Figure 5, by clicking the **Postavi** (Set) button, the read data is stored in the database. With this, one Evaluation Sheet was read and stored in the database. Also, by clicking on the button, the file, in which the corrected evaluation sheet is stored, is saved, named student ID with the extension .jpeg.

3.2. Problems when reading the Evaluation Sheets

There are several challenges that can arise during the reading of Evaluation Sheets, resulting in incomplete or inaccurate data extraction. This is especially true for failing to read the Student ID or First Foreign Language ID or Test Combination ID. The causes can be various, but the most common are:

- Adding unnecessary marks to the Evaluation Sheet by the student
- Greasy fingers from breakfast or food stains on the Evaluation Sheets.
- Transferring makeup or other substances onto the Evaluation Sheets, creating prints or stains.
- Excessively slanted or reverse scanning (rating sheet turned 180 degrees during scanning) of grading sheets.

The first reason could be solved by better preparation of the students for the matriculation exam, i.e. by filling in the evaluation sheets with the note that nothing else should be added to the Evaluation Sheet except to transfer their answers from the Integral Test to the Evaluation Sheet.

The second and third reasons could be prevented by better preparation of the teachers on duty who conduct the External Matura in a certain elementary school.

The fourth reason can be prevented by better preparation of Evaluation Sheets before the start of their scanning, as well as by purchasing better quality paper, at least for Evaluation Sheets, as well as by purchasing better quality scanners. In the previously mentioned cases when the Evaluation Sheet is not read in its entirety, then we have two cases:

- If it is a semi-automatic reading, then the user can manually enter the missing data by visual inspection of the Evaluation Sheet (Figure 5 - Review of the Evaluation Sheet) and then store the data from that Evaluation Sheet in the database via the button **Postavi** (Set).
- If it is an automatic reading, then for the Evaluation Sheets that are not read completely in the automatic reading interface, a message that the evaluation sheet was not read is printed with a description of the log, and it continues to read the next Evaluation Sheet. After the auto reader has finished its process of reading all the mark sheets e.g. for one school, then semi-automatic reading is applied to mark sheets that have not been successfully read.

In addition to the aforementioned controls, a certain number of logical controls were added to the software system in the sense that the student ID must comply with certain rules such as length, range in which the student ID can move, etc. Overall, the system has very good filters so that it is almost impossible for a grade sheet to enter the system with a student ID that does not belong to any student taking the matriculation exam, or that the first foreign language as well as the ID of the test combination has not been identified.

3.3 Scoring of Evaluation Sheets

Scoring of the Evaluation Sheets is done in such a way that the stored answers from the Evaluation Sheet are compared with the correct answers in the database and one point is added for each correct answer. After scoring all the questions (60), the total number of points is obtained. Based on the total number of points, a percentage of the maximum number of points is calculated, which represents the total result of the matriculation exam (testing) for one student.

When scoring the answers to the questions, the software system, based on the number and structure of questions in certain areas (subjects) shown in Table 1, calculates the total number of points per area (subjects), and then based on that, the percentage of the maximum number of points in a certain area (subject) is calculated field, which represents the result of taking the matriculation exam in a particular subject.

Figure 10 shows the interface for evaluating (scoring) stored evaluation sheets.

Figure 10. Interface for scoring (evaluation) of stored Evaluation Sheets

Scoring of Evaluation Sheets can be done in two ways:

- Individually for one student by entering the Student ID in the Student Code (Šifra učenika) field and clicking the **Bodovanje** (Scoring) button
- For all students of one school by entering the school ID in the School Code (Šifra škole) field and clicking on the **Bodovanje** (Scoring) button

Within the scoring interface (Figure 10) there is a **chk** field with the label “Ispiši rezultat na ocjenjivačke listove” (Print the result on the Evaluation Sheets), when included during scoring in the student's Evaluation Sheet, prints the overall test result, as well as the results by areas (subjects). Figure 11 shows an Evaluation Sheet that was scored in the previously described manner. All scoring results of the Evaluation Sheets are stored in the database.

Subject	Score	Percentage
1. Bosanski, hrvatski, srpski jezik i književnost (1-5):	5	100,00%
2. Engleski jezik - prvi strani jezik (6-10):	5	100,00%
3. Matematika (11-15):	5	100,00%
4. Fizika (16-20):	5	100,00%
5. Informatika (21-25):	4	80,00%
6. Biologija (26-30):	5	100,00%
7. Hemija/Kemija (31-35):	5	100,00%
8. Historija/Povijest (36-40):	5	100,00%
9. Geografija/Zemljopis (41-45):	5	100,00%
10. Tehnička kultura (46-49):	2	66,67%
11. Likovna kultura (50-53):	3	100,00%
12. Muzička kultura (54-57):	3	100,00%
13. Tjelesni i zdravstveni odgoj (58-59):	3	100,00%
14. Građansko obrazovanje (60):	2	66,67%

Figure 11. Scored Evaluation Sheet for one student

Figure 12 shows how the software system printed the test results by area (subjects) as well as the total test result for one student (surname and first name of the student are covered) on the Evaluation Sheet.



Figure 12. Printout of test results by areas (subjects) and in total

Likewise, next to each question, the software system puts a mark on whether the transmitted answer is correct or not, and along with that mark, it also puts the correct answer to that question for that test combination as it is shown in Figure 13.

	A	B	C	D	
21.	☐	☐	☐	☒	21. ✓ D
22.	☐	☒	☐	☐	22. ✓ B
23.	☒	☐	☐	☐	23. ✓ A
24.	☐	☐	☐	☒	24. ✓ D
25.	☐	☐	☒	☐	25. ✗ A
26.	☒	☐	☐	☐	26. ✓ A

Figure 13. Print marks for correct or incorrect answers to questions

In Figure 13, it can be seen that all the answers were correctly read and evaluated, so only the 25th question was incorrectly answered with the mark C, and the correct answer was under A.

4. Conclusion

The implementation of the Software System for automatic reading, storing, and evaluation of scanned Evaluation Sheets in the field of education has been successfully developed with the use of various techniques of reading and recording marked answers, as well as other test data. In the last school year, about 4,000 assessment sheets were examined and evaluated within the framework of matriculation examinations in elementary schools of Tuzla Canton with the help of the described software system. Following the publication of test results, the scoring Evaluation Sheets, along with the corresponding test combination, were electronically provided to the students for review. While the system generally yielded satisfactory results, there was one student complaint regarding the interpretation of transferred answers on the Evaluation Sheet. The student had misunderstood the logic for canceling answers, marking the desired answer as 'Cancelled' and the incorrect answer as 'Correct'. Such instances provided insights into potential improvements for the system. In the initial years of application, a semi-automatic model of reading and scoring Evaluation Sheets was more used, through which the possibilities of improving the system itself were recognized. In recent years, as confidence in the results of the review of Evaluation Sheets has grown, the automatic model has been used more. Although the

presented system was created for questions with the choice of one correct answer from several offered answers, it can be upgraded with very little effort for questions with the choice of more than one correct answer from several offered answers. Given that the developed system was not created for commercial use, but for internal use in the process of passing the matriculation exam for elementary school students, not much work was done on the appearance of the software interface itself. Of course, in future upgrades of this system, work will be done on improving the user interface as well as adding new features that will aim to increase the flexibility and capabilities of the software system.

References

- [1] Awny Abbas A 2009 An automatic system to grade multiple choice questions paper based exams *Journal of University of Anbar for Pure Science* **3** 174–81 doi: 10.37652/juaps.2009.15510
- [2] Nicol D 2007 E-assessment by design: Using multiple-choice tests to good effect *Journal of Further and Higher Education* **31** 53–64 doi: 10.1080/03098770601167922
- [3] Afifi M and Hussain K F 2019 The achievement of higher flexibility in multiple-choice-based tests using image classification techniques *International Journal on Document Analysis and Recognition (IJDAR)* **22** 127–42 doi: 10.1007/s10032-019-00322-3
- [4] Gronlund N E 1998 *Assessment of student achievement*. Sixth edition
- [5] McCoubrie P 2004 Improving the fairness of multiple-choice questions: A literature review *Medical Teacher* **26** 709–12 doi: 10.1080/01421590400013495
- [6] Kommey B, Keelson E, Samuel F, Twum-Asare S and Akuffo K K 2022 Automatic Multiple Choice Examination questions marking and grade generator software IPTEK The Journal for Technology and Science **33** 175 doi: 10.12962/j20882033.v33i3.14522
- [7] Hafeez Q, Aslam W, Ikramullah Lali M, Ahmad S, Alqahtani M and Shafiq M 2023 Fault tolerant optical mark recognition *Computers, Materials & Continua* **74** 3829–47 doi: 10.32604/cmc.2023.026422
- [8] Nguyen T D, Hoang Manh Q, Bui Minh P, Nguyen Thanh L and Hoang T M 2011 Efficient and reliable camera based multiple-choice test grading system The 2011 International Conference on Advanced Technologies for Communications (ATC 2011) doi: 10.1109/atc.2011.6027482